

CAROLLINE: A Caged Rolling Drone With Omnidirectional Ground Mobility and Simple Implementation

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Abstract—In this letter, we propose modeling and control design of a caged rolling drone, called CAROLLINE, that is capable of both ground and aerial locomotion with simple implementation. Specifically, omnidirectional ground mobility is achieved without additional actuators or devices, which contributes to miniaturization and operational efficiency of CAROLLINE. A mathematical model of CAROLLINE is constructed to design an appropriate controller for holonomic locomotion and to assume a pre-takeoff posture from any initial attitude. The overall control scheme is modularly composed and integrated well with the existing drone system. The proposed rolling on the ground provides 12.6 times higher energy efficiency than conventional flight. Experimental validation was also conducted to demonstrate the actual performance of CAROLLINE.

Index Terms—Aerial systems: mechanics and control, field robots, actuation and joint mechanisms.

I. INTRODUCTION

MULTIROTORS, featuring a simple structure and flexibility in movement such as vertical takeoff and landing

Received 16 April 2025; accepted 29 August 2025. Date of publication 10 September 2025; date of current version 15 September 2025. This article was recommended for publication by Associate Editor J. Gangloff and Editor C. Gosselin upon evaluation of the reviewers' comments. This work was supported in part by the Institute of Information and Communications Technology Planning and Evaluation (IITP) funded by the Korea Government Ministry of Science and ICT (MSIT) under Grant 2019-0-00762, in part by the development of sensor and robot technology to support life search and fire suppression activities at firefighting sites program of the National Fire Agency under Project 20026197, Development of track-based mobile robot for fire suppression and development of firefighting environment demonstration technology, in part by the 2024 Cultural Heritage Smart Preservation and Utilization Technology Development Program of the Cultural Heritage Administration and the National Research Institute of Cultural Heritage (Program Title: Development of software intelligent solution for digital archaeological drawing), and in part by MSIT through the ITRC (Information Technology Research Center) Support Program under Grant IITP-2025-RS-2024-00438430 supervised by IITP. (*Corresponding author : Soohye Han.*)

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Demonstration video: <https://youtu.be/sXpN0pgBLx4>.

This article has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2025.3608651>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2025.3608651

(VTOL) and hovering, have been widely employed in various fields including indoor exploration, logistics management, factory inspection, environmental monitoring, and disaster relief [1], [2], [3]. However, the short operational time caused by low energy efficiency, and the safety issues owing to vulnerability to collisions during flight remain major challenges that must be addressed for actual deployment in industrial settings [4], [5], [6].

A hybrid flying and driving locomotion, achieving the agility of aerial motion with the high energy efficiency on ground locomotion has attracted significant interest in the robotics community as a promising method to overcome the limited operational time of conventional multirotors [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Flying STAR [7], which is equipped with propellers for flight and wheels for ground motion, utilizes an additional actuator for sprawl motion. Furthermore, to simplify its structure, researchers attempted to share actuators for propellers and wheels to enable a single motor to handle both [8]. As one approach to address the limitations of wheel-based ground travel, the walking mechanism has also been studied. For instance, [9] combined a hexacopter with six legs driven by 18 motors, whereas other research has integrated multirotors with quadruped [10] or humanoid [11] walking. Additionally, [12] attached extra actuators to a conventional propeller-wheel-based hybrid locomotion robot, offering additional degrees of freedom and enabling more versatile shape-shifting capabilities for robots. This design evolved into the Multi-Modal Mobility Morphobot M4, which can perform flying, ground driving, crawling, crouching, balancing, tumbling, and walking. However, these vehicle systems require numerous transmissions or actuators to implement hybrid locomotion. This not only complicates the manufacturing process and makes it difficult to attain a compact size and low weight, limiting operation in complex or confined environments, but also requires more actuators to be operated, increasing power consumption and reducing energy efficiency. Moreover, this complex system structure makes controller design more challenging. In addition, safety risks remain unless in-flight collision problems are directly addressed.

For simple implementation, protective cages have been employed. They facilitate structural simplicity by minimizing additional mechanical components, while ensuring safety in

the event of collisions and enhancing energy efficiency in ground mobility, thus offering high practicality in congested and cluttered environments. As a basic attempt, a cylindrical cage was attached to a drone using only a single-axis revolute joint [13], [14]. The drone's thrust was transformed into rotational motion of the cage through the joint, enabling terrestrial locomotion. Alternatively, instead of adding external parts to couple the cage with the drone, a system was proposed that achieves rolling by adjusting the propeller thrust directions using onboard mechanical devices [15], [16]. However, in these methods, the motion guide through a one-dimensional revolute joint and the cage's inherent cylindrical structure do not structurally allow movement in all rolling directions. Although three-axis gimbals, variable-pitch propellers or additional propellers have been introduced to provide omnidirectional terrestrial locomotion for multirotors with protective cages [17], [18], [19], as previously mentioned, the introduction of complex mechanical components hinders compact and lightweight designs, maneuverability in tight environments, and efficient energy consumption.

To overcome the aforementioned problems, this paper proposes the design, modeling, and control of "CAROLLINE," a structurally simple spherical-cage-equipped hybrid drone that can perform omnidirectional terrestrial locomotion without requiring additional mechanical devices or actuators. CAROLLINE rigidly couples a typical drone with a spherical cage, thereby achieving holonomic ground motion without requiring any extra coupling device. To achieve this, we present a driving method leveraging the bidirectional operation of a single motor along with corresponding modeling and control techniques. CAROLLINE can create thrust in both upward and downward directions by operating a single motor in a bidirectional manner. This type of thrust enables CAROLLINE to actively adjust the rolling of its cage and thereby achieve omnidirectional holonomic motion on the ground. We also mathematically model the interaction between the cage's rolling motion and the thrust vectors to systematically derive the control inputs required for CAROLLINE's omnidirectional ground mobility. In addition, a pre-takeoff controller is proposed to ensure that the appropriate thrust vector for takeoff can be generated at any attitude during ground movement. Real-world ground experiments confirmed that the proposed control scheme can achieve CAROLLINE's omnidirectional locomotion, and also verified the feasibility of quickly switching to a pre-takeoff mode from any arbitrary orientation. Finally, to confirm the energy efficiency of CAROLLINE, we compare the energy consumption of ground and flight locomotion over the same trajectory, showing that the former is approximately 12.6 times more energy efficient than the latter.

The key contributions of this work can be summarized as follows:

- *Omnidirectional ground mobility and simple implementation:* We propose a hardware configuration for omnidirectional rolling using only the bidirectional rotation of a single motor-propeller system, without additional mechanical components. This design provides a solid foundation for lightweight, compact cage-based hybrid drone

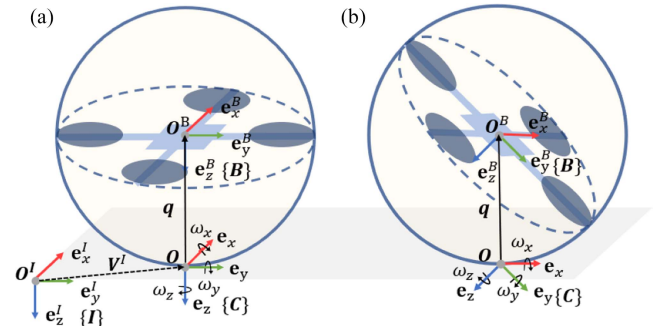


Fig. 1. The inertial and body frames, and the contact point frame used to describe the rolling motion of CAROLLINE. (a) Horizontal position (b) Tilted position.

systems and can be effectively applied in confined indoor environments.

- *Modeling and control design for omnidirectional movement and takeoff behavior:* We model the linear velocity resulting from the cage's rolling motion and design a controller that uses the drone's dynamics to achieve omnidirectional holonomic movement. Furthermore, we devise a pre-takeoff mode-switching controller to enable immediate transition to flight mode before taking off from the ground.
- *Experimental validation for control performance and energy consumption:* We verified the efficacy and performance of the proposed controller through ground trajectory tracking and pre-takeoff experiments. A comparison of energy consumption between ground travel and flight along the same path revealed that the proposed ground locomotion offers about 12.6 times higher energy efficiency than flight.

Section II presents the mathematical modeling of the roadable CAROLLINE. Section III provides details on the rolling controller for generating the desired translational velocity, along with the pre-takeoff controller that assumes the attitude suitable for takeoff. Next, Section IV describes the detailed implementation, followed by Section V, which reports the experimental results validating the proposed model and controller, including a discussion of energy efficiency. The final Section VI concludes the paper and outlines potential directions for future research.

II. MODELING

In this section, the rolling motion of CAROLLINE is mathematically modeled and utilized in the next section to design the controller. CAROLLINE consists of a drone rigidly mounted inside a spherical cage. This motion occurs from the motors attached to the drone. Here, we present the corresponding mathematical model in three parts: *Rolling Kinematics*, *Time-Varying Inertia Tensor*, and *Drone Dynamics*.

Fig. 1 depicts three frames required to describe CAROLLINE's motion. The fixed inertial frame $\{I\}$, in which Newton's first law holds, has bases $\{e_x^I, e_y^I, e_z^I\}$, and the origin O^I . The body frame $\{B\}$, fixed to the drone, has bases $\{e_x^B, e_y^B, e_z^B\}$ and its origin O^B coincides with the drone's center of mass. The

contact point frame $\{C\}$, located at the contact point O , has, as its bases, the instantaneous axis of rotation (IAR) $\{e_x, e_y, e_z\}$ of the spherical cage. The reference frame is indicated in the superscript; if not explicitly specified, it is assumed to be expressed in the contact point frame $\{C\}$. Hence, the angular velocity about the IAR is denoted by $\omega = [\omega_x, \omega_y, \omega_z]^T \in \mathbb{R}^3$. The rotation directions follow the right-hand rule, where counterclockwise rotations about each axis are considered positive.

As shown in Fig. 1, when the rigid sphere rotates, the IAR is based on the contact point between the cage and ground, and it rotates accordingly as the drone rolls. $\mathbf{R} \in SO(3)$ represents the rotation from the inertial frame $\{I\}$ to the contact point frame $\{C\}$, where

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}\mathbf{R}^T = \mathbf{E}, \det(\mathbf{R}) = 1\}, \quad (1)$$

and $\mathbf{E} \in \mathbb{R}^{3 \times 3}$ is the identity matrix. Although $\mathbf{R}$ is time-varying, the time variable t is omitted for brevity. While CAROLLINE is rolling, the axis vectors of the contact frame are expressed in the inertial frame as

$$\mathbf{e}_i = \mathbf{R}\mathbf{e}_i^I, \quad \text{for } i \in \{x, y, z\}. \quad (2)$$

A. Rolling Kinematics

When CAROLLINE rolls on a flat surface, it exhibits the typical characteristics of a spherical mobile robot [20]. In this context, its angular velocity ω about the IAR can be decomposed into independent rotational components ω_x, ω_y , and ω_z along mutually orthogonal axes. Each rotational component generates a corresponding linear velocity on the ground. The rolling kinematics describe the relationship between this angular velocity and the resulting linear velocity.

The robot's linear velocity $\mathbf{V}^I = [V_x^I, V_y^I]^T \in \mathbb{R}^2$ is measured in O with respect to the inertial frame $\{I\}$. The Jacobian matrix $\mathbf{J} \in \mathbb{R}^{2 \times 3}$ maps the linear velocity $\mathbf{V}^I$ to the angular velocity ω :

$$\mathbf{V}^I = \mathbf{J}\omega, \quad (3)$$

where $\mathbf{J}$ is computed from

$$\begin{aligned} \mathbf{J} &= \begin{bmatrix} J_x & J_y & J_z \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{e}_x \times \mathbf{q}^I & \mathbf{e}_y \times \mathbf{q}^I & \mathbf{e}_z \times \mathbf{q}^I \end{bmatrix}_{1:2}, \end{aligned} \quad (4)$$

and $\mathbf{q}^I \in \mathbb{R}^3$ denotes the distance vector from O to O^B . Note that only the first two rows in (4) are used because we consider only 2D motion on the ground. This interpretation is based on previous work verifying the controllability of a spherical robot using three independent disc actuators [21].

B. Time-Varying Inertia Tensor

To describe the moments generated by rotational motion, we must reflect changes in the inertia tensor due to the varying IAR. Therefore, we compute the time-varying inertia tensor $\mathcal{I} \in \mathbb{R}^{3 \times 3}$ depending on the rotation of the drone. The moment of inertia of the rigid body about the center of mass O^B , denoted by $\mathcal{I}^B \in$

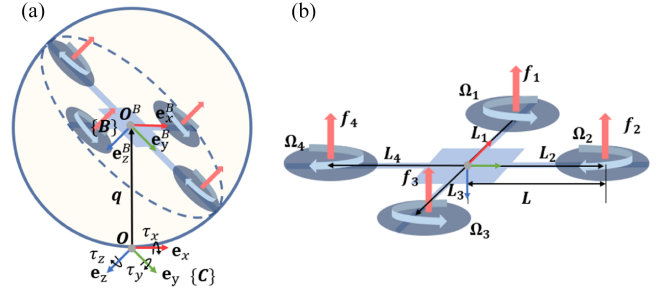


Fig. 2. (a) Directions of the thrust and torque generated by the drone's propellers during rolling motion. (b) Physical variables and parameters relevant to drone dynamics.

$\mathbb{R}^{3 \times 3}$, is given by

$$\mathcal{I}^B = \begin{bmatrix} \mathcal{I}_{xx}^B & -\mathcal{I}_{xy}^B & -\mathcal{I}_{xz}^B \\ -\mathcal{I}_{yx}^B & \mathcal{I}_{yy}^B & -\mathcal{I}_{yz}^B \\ -\mathcal{I}_{zx}^B & -\mathcal{I}_{zy}^B & \mathcal{I}_{zz}^B \end{bmatrix}. \quad (5)$$

However, as mentioned earlier, CAROLLINE's IAR is located at the contact point on the ground, and the rotation axis changes with time when rolling. Hence, we must transform the moment of inertia for rotation about the IAR using the parallel axis theorem, as follows:

$$\mathcal{I} = \mathcal{I}^B + M[(\mathbf{q}^T \mathbf{q})\mathbf{E} - \mathbf{q}\mathbf{q}^T], \quad (6)$$

where $\mathcal{I}^B$ is the moment of inertia measured at O^B , M is the total mass of the robot. The tensor $\mathcal{I}$ obtained from (6) comprises the inertia values corresponding to rotational motion about the IAR. Note that the mass of the outer cage is included in M .

C. Drone Dynamics

The drone is mounted inside the cage, as shown in Fig. 2(b). It has four motors with propellers: motors 1 and 3 spin clockwise, whereas motors 2 and 4 spin counterclockwise. The spinning motors produce a thrust vector $\mathbf{F} = [f_1, f_2, f_3, f_4]^T \in \mathbb{R}^4$ and a corresponding drag-induced torque vector $\boldsymbol{\Omega} = [\Omega_1, \Omega_2, \Omega_3, \Omega_4]^T \in \mathbb{R}^4$. The motor torque and thrust is linearly dependent on each other, and can be expressed using a scale factor μ [22], [23], [24], [25]:

$$\boldsymbol{\Omega} = \mu \mathbf{F}. \quad (7)$$

The spinning of CAROLLINE's motors generates a rolling torque $\boldsymbol{\tau} = [\tau_x, \tau_y, \tau_z]^T \in \mathbb{R}^3$ about the IAR. Due to the geometric structure of CAROLLINE, the thrust force $\mathbf{F}$ governs the motion about the $\mathbf{e}_x$ and $\mathbf{e}_y$ axes, while the drag-induced torque $\boldsymbol{\Omega}$ drives the motion about the $\mathbf{e}_z$ axis. Consequently, the rolling torque can be separated into $[\tau_x, \tau_y]^T \in \mathbb{R}^2$, generated by the thrust, and τ_z , arising from the drag-induced torque. Thrust forces are generally much larger than drag-induced torques. The rolling torques due to thrust is given by:

$$\begin{bmatrix} \tau_x \\ \tau_y \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^4 ((\mathbf{q} + \mathbf{L}_j) \times f_j(-\mathbf{e}_z)) \end{bmatrix}_{1:2}, \quad (8)$$

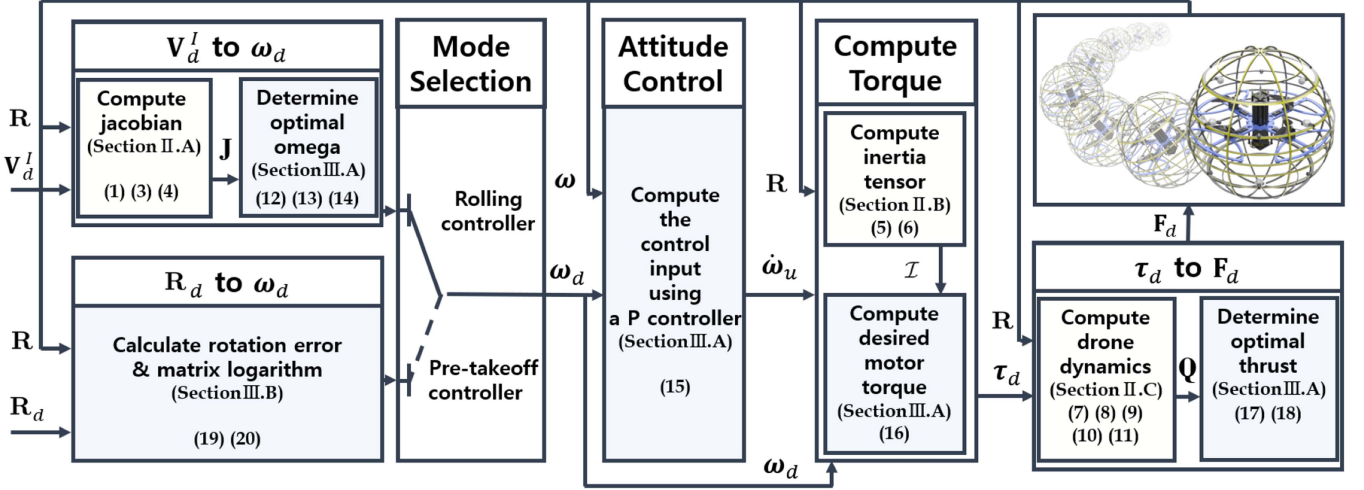


Fig. 3. Overview of CAROLLINE's controller. The numbers in parentheses within the box represent the relevant equations.

where $\mathbf{L}_j$ is the distance vector from the drone's center to the j^{th} propeller. The first, second, third, and fourth propellers are located along the $\mathbf{e}_x$, $\mathbf{e}_y$, $-\mathbf{e}_x$, and $-\mathbf{e}_y$ directions, respectively. They are all separated by distance $L \in \mathbb{R}$ from the center of the drone. The thrust $f_j \in \mathbb{R}$ from each propeller is in the direction of $-\mathbf{e}_z$.

The rolling torque about the $\mathbf{e}_z$ axis can be obtained as follows:

$$\begin{aligned} \tau_z &= -\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4 \\ &= -\mu f_1 + \mu f_2 - \mu f_3 + \mu f_4. \end{aligned} \quad (9)$$

By combining (8) and (9), we define the matrix $\mathbf{Q} \in \mathbb{R}^{3 \times 4}$ that maps the thrust vector $\mathbf{F}$ to the rolling torque $\boldsymbol{\tau}$:

$$\boldsymbol{\tau} = \mathbf{Q} \mathbf{F}, \quad (10)$$

where q is the sphere's radius, $\mathbf{R}_{31}$, $\mathbf{R}_{32}$ are the (3,1) and (3,2) entries of the rotation matrix $\mathbf{R}$, respectively, and $\mathbf{Q}$ is defined by

$$\mathbf{Q} = \begin{bmatrix} q \mathbf{R}_{32} & q \mathbf{R}_{32} - L & q \mathbf{R}_{32} & q \mathbf{R}_{32} + L \\ -q \mathbf{R}_{31} + L & -q \mathbf{R}_{31} & -q \mathbf{R}_{31} - L & -q \mathbf{R}_{31} \\ -\mu & \mu & -\mu & \mu \end{bmatrix}. \quad (11)$$

In the next section, the mathematical models in this section serve as fundamental components for constructing CAROLLINE's controller.

III. CONTROL DESIGN

In this section, we design the tracking controller based on the mathematical models described in the previous section. As shown in Fig. 3, the controller has two modes, i.e., rolling and pre-takeoff modes. Each mode employs a cascaded controller structure that sequentially uses attitude control, compute torque, and $\boldsymbol{\tau}_d$ to $\mathbf{F}_d$ modules. The difference between two modes lies in how the desired angular velocity $\boldsymbol{\omega}_d = [\omega_{dx}, \omega_{dy}, \omega_{dz}]^T \in$

$\mathbb{R}^3$ is obtained, either using the $\mathbf{V}_d^I$ to $\boldsymbol{\omega}_d$ module or the $\mathbf{R}_d$ to $\boldsymbol{\omega}_d$ module. The mode is selected according to operational requirements. Next, each mode's controller will be described in detail.

A. Rolling Controller

The rolling controller aims to achieve the desired linear velocity

$$\mathbf{V}_d^I = \begin{bmatrix} V_{dx}^I \\ V_{dy}^I \end{bmatrix} \in \mathbb{R}^2$$

for terrestrial locomotion. Using the Jacobian (4), which reflects the current attitude, the controller computes the corresponding angular velocity $\boldsymbol{\omega}_d$. The matrix $\mathbf{J}$ is not square, so it is noninvertible. Consequently, no unique angular velocity solution exists for a given $\mathbf{V}_d^I$. We seek an optimal solution under certain constraints.

As mentioned earlier, the components ω_x and ω_y , corresponding to rotations about the axes $\mathbf{e}_x$ and $\mathbf{e}_y$, are directly actuated by motor thrust. In contrast, ω_z , the angular velocity about the axis $\mathbf{e}_z$, is generated solely by the drag-induced torque, which is inherently much weaker than the thrust. This difference makes thrust-induced motion more advantageous than motion arising from drag-induced torque. In this regard, the optimization process is employed to reduce dependency on the rotational motion about the $\mathbf{e}_z$ axis. Hence, we consider the following optimization problem:

$$\min_{\boldsymbol{\omega}} \frac{1}{2} \boldsymbol{\omega}^T \mathbf{W}_{\omega} \boldsymbol{\omega} \quad \text{subject to} \quad \mathbf{V}_d^I = \mathbf{J} \boldsymbol{\omega}, \quad (12)$$

where $\mathbf{W}_{\omega} \in \mathbb{R}^{3 \times 3}$ is a positive semidefinite weight matrix. The diagonal components W_x , W_y , and W_z in $\mathbf{W}_{\omega}$ represent the weights associated with rotational axes:

$$\mathbf{W}_{\omega} = \begin{bmatrix} W_x & 0 & 0 \\ 0 & W_y & 0 \\ 0 & 0 & W_z \end{bmatrix}, \quad (13)$$

where a larger W_z reduces reliance on the rotational motion about the $\mathbf{e}_z$ axis. Solving the constrained optimization problem (12), we have the following optimal angular velocity:

$$\boldsymbol{\omega}_d = \mathbf{W}_\omega^{-1} \mathbf{J}^\top \left(\mathbf{J} \mathbf{W}_\omega^{-1} \mathbf{J}^\top \right)^{-1} \mathbf{V}_d^I. \quad (14)$$

Because the attitude is measured at a high sampling rate, we assume that the accumulated error remains negligible, thus adopting a simple proportional controller without an integral term:

$$\dot{\boldsymbol{\omega}}_u = K_p (\boldsymbol{\omega}_d - \boldsymbol{\omega}), \quad (15)$$

where $K_p \in \mathbb{R}^{3 \times 3}$ is a diagonal gain matrix with positive scalar entries, and $\boldsymbol{\omega} \in \mathbb{R}^3$ is the current angular velocity vector.

The desired angular acceleration input $\dot{\boldsymbol{\omega}}_u \in \mathbb{R}^3$ is converted into a rolling torque via the rigid body's varying inertia. Thus, the desired torque

$$\boldsymbol{\tau}_d = [\tau_{dx}, \tau_{dy}, \tau_{dz}]^\top \in \mathbb{R}^3$$

accounting for CAROLLINE's time-varying inertia tensor $\mathcal{I}$ is computed as

$$\boldsymbol{\tau}_d = \mathcal{I} \dot{\boldsymbol{\omega}}_u + \boldsymbol{\omega}_d \times (\mathcal{I} \boldsymbol{\omega}_d). \quad (16)$$

To obtain the total torque, we also need to consider rolling resistance and drag force in the direction of the current $\boldsymbol{\omega}$; however, these are omitted here for brevity.

Finally, to generate the desired rolling torque, we compute the motor thrust vector

$$\mathbf{F}_d = [f_{1d}, f_{2d}, f_{3d}, f_{4d}]^\top \in \mathbb{R}^4,$$

using the mapping matrix $\mathbf{Q} \in \mathbb{R}^{3 \times 4}$ defined in (10) and (11). Because $\mathbf{Q}$ is non-invertible, infinitely many solutions exist for $\boldsymbol{\tau}_d = \mathbf{Q} \mathbf{F}$. Among these, we select the set of thrusts that minimizes the total effort:

$$\min_{\mathbf{F}} \frac{1}{2} \mathbf{F}^\top \mathbf{F} \quad \text{subject to} \quad \boldsymbol{\tau}_d = \mathbf{Q} \mathbf{F}. \quad (17)$$

Applying the stationary condition yields the optimal thrust:

$$\mathbf{F}_d = \mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top)^{-1} \boldsymbol{\tau}_d. \quad (18)$$

By commanding the drone motors to produce this thrust, CAROLLINE achieves the desired velocity $\mathbf{V}_d^I$.

B. Pre-Takeoff Controller

When CAROLLINE transitions from terrestrial locomotion to flight, all propellers must point skyward. If the body frame is arbitrarily oriented, manually aligning the propellers becomes difficult. Therefore, we introduce a pre-takeoff mode. In this mode, the $\mathbf{R}_d$ to $\boldsymbol{\omega}_d$ module uses the matrix logarithm to compute the angular velocity $[\boldsymbol{\omega}_d] \in \mathfrak{so}(3)$ that moves the current attitude $\mathbf{R}$ to $\mathbf{R}_d$ [26], [27]:

$$[\boldsymbol{\omega}_d] = \log(\mathbf{R}^\top \mathbf{R}_d). \quad (19)$$

The skew-symmetric matrix $[\boldsymbol{\omega}_d]$ is expressed as

$$[\boldsymbol{\omega}_d] = \begin{bmatrix} 0 & -\omega_{dz} & \omega_{dy} \\ \omega_{dz} & 0 & -\omega_{dx} \\ -\omega_{dy} & \omega_{dx} & 0 \end{bmatrix}. \quad (20)$$

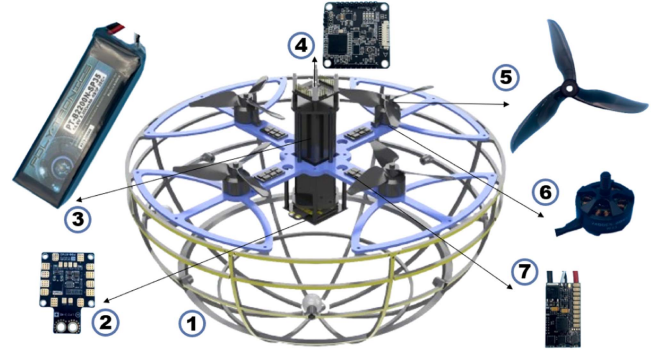


Fig. 4. Composition of CAROLLINE's hardware: (1) Spherical Cage, (2) Power Distribution Board, (3) Lithium-Polymer Battery, (4) Flight Controller, (5) Propeller, (6) Brushless Motor, (7) Electronic Speed Controller.

Its isomorphic vector form $\boldsymbol{\omega}_d$ represents the angular velocity required for the drone to achieve the desired attitude. By computing the difference in attitude directly on the manifold, this module determines a short path to reach $\mathbf{R}_d$.

Subsequently, we follow the same procedure used in the rolling controller, (15)–(18), to generate the thrust that produces the required angular velocity $\boldsymbol{\omega}_d$.

IV. IMPLEMENTATION

In this section, we provide details on CAROLLINE's design and the experimental environment, as shown in Fig. 4. To ensure algorithmic generality, our objective was to enable terrestrial motion on a stereotypical drone with minimal modifications.

A. Hardware Design

CAROLLINE comprises a drone and spherical cage. The drone follows a common form described in previous works [24]. Unlike previous designs that use a revolute joint to rotate the cage [13], [17], CAROLLINE has no transmission mechanism. Instead, its unique structural feature is the vertical “pillar” placement of the battery, which aids in balancing the inertia. As explained earlier, rolling torque in the $\mathbf{e}_x$ and $\mathbf{e}_y$ directions is generated by the motor thrust $\mathbf{F}$, whereas that in the $\mathbf{e}_z$ direction is generated by the motor torque Ω . By orienting the battery in a pillar-like configuration, we nearly equalize the inertias about $\mathbf{e}_x^B$ and $\mathbf{e}_y^B$. In the original design, the difference in these two inertias was approximately $3.04 \times 10^5 \text{ g} \cdot \text{mm}$, whereas our modification reduces the difference to $9.53 \times 10^4 \text{ g} \cdot \text{mm}$. Furthermore, this arrangement reduces the inertia about the $\mathbf{e}_z^B$ axis by $1.72 \times 10^5 \text{ g} \cdot \text{mm}$, thereby lowering the rolling torque required to achieve the same angular speed around $\mathbf{e}_z^B$ —an advantage given that the torque in this axis is comparatively weaker.

Because rolling is produced by body-mounted motor thrusts, the cage must be rigidly fixed to the drone. The cage design is based on the spherical drone in [28], but with additional ribs to ensure it is nearly spherical for smooth rolling. We fabricated both the body and cage via 3D printing. Table I presents key specifications of the robot used in our experiments.

TABLE I
PHYSICAL PARAMETERS

Symbol	Description and Value
M	Mass: 0.937 kg
L	Arm length: 0.1125 m
q	Sphere radius: 0.2 m
μ	Conversion coefficient: 1.54×10^{-2} m
$\mathcal{I}_{xx}$	x -axis Inertia: $1.08 \times 10^9 \text{ g} \cdot \text{mm}^2$
$\mathcal{I}_{yy}$	y -axis Inertia: $1.08 \times 10^9 \text{ g} \cdot \text{mm}^2$
$\mathcal{I}_{zz}$	z -axis Inertia: $1.32 \times 10^9 \text{ g} \cdot \text{mm}^2$

B. Experimental Equipment

CAROLLINE's electronic components are powered by a 14.8 V, 2200 mAh, 4S lithium-polymer battery. To supply power to the four propulsion modules and the flight controller (FC), we used a MATEKSYS PDB-XT60 power distribution board. The control algorithm runs on a custom-made FC equipped with an STM32 F407 microcontroller and an LSM9DS0 inertial measurement unit. The microcontroller collects the data necessary for control and also handles wireless communication with a ground control unit. Our experimental setup includes a motion capture system that detects the infrared (IR) markers attached to the robot and computes its position and attitude. These measurements are fused with the data from the FC's onboard IMU. Including the data acquisition via wireless communication, the controller application is implemented at a sampling time of 4 ms.

C. Propulsion Module

The final step of the cascaded controller is to have the drone motors produce thrust. This is achieved by CAROLLINE's propulsion module, which is composed of four off-the-shelf C2206–2300KV brushless motors, three-blade CYCLONE T5040 Pro propellers, and AK32 35 A electronic speed controllers (ESCs). Unlike a conventional drone—whose thrust is applied in only one direction, CAROLLINE produces torque in both the positive and negative directions for omnidirectional rolling. In other words, its propellers operate to produce bidirectional thrust. To do this, we apply the BLHeliSuite32 program to set the ESCs to the “Bidirectional Soft” mode, with “Rampup Power” at 100%. As a result, when the ESC signal deviates negatively from 1500 (down to 1040), the motors spin in reverse, and when it increases positively (up to 1960), they spin in the forward direction. This method is adopted from the omnicopter build manual [29].

The aforementioned bidirectional setup can result in reduced thrust during the reverse spin of motors. To address this issue, one potential solution is to use bidirectional propellers specifically designed to produce nearly symmetric thrust in both directions [30]. However, to minimize hardware modifications and ensure compatibility with standard drone components, CAROLLINE uses off-the-shelf unidirectional propellers. In this work, the thrust asymmetry caused by unidirectional propellers is addressed by incorporating bidirectional thrust characteristics into the thrust command generation.

For precise control, we must determine how the ESC signal correlates with the thrust produced by the propellers when

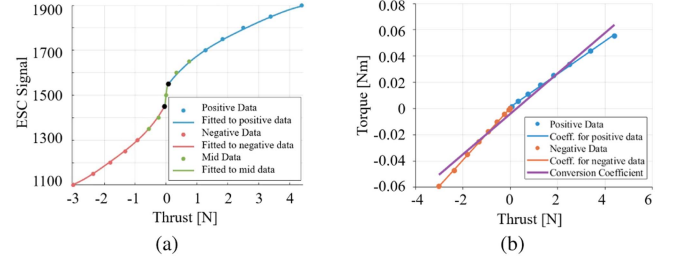


Fig. 5. (a) Polynomial fit from thrust to ESC signal. (b) Conversion coefficient μ from thrust to torque.

rotating in both positive and negative directions. Hence, we connected the configured ESC, along with a motor and propeller of the drone, to a force sensor (TYTO ROBOTICS Dynamometer Series 1580). We varied the ESC signal from 1100 to 1900 in increments of 50 and recorded the resulting torque and thrust. As shown in Fig. 5(a), the relationship between the ESC signal and thrust data were approximated using polynomials. Because the propeller configurations differ for negative, mid-range, and positive signal regions, we performed separate fitting for these three segments. To ensure a one-to-one correspondence between thrust and signal, we constrained each polynomial to have the same slope and function value at ESC signals of 1550 and 1450. Additionally, using the measured torque and thrust data from this setup, we applied a linear regression to determine the conversion coefficient μ , as illustrated in Fig. 5(b).

V. EXPERIMENTAL VALIDATION

We validated the performance of CAROLLINE's rolling controller by conducting experiments of tracking a predefined path. Furthermore, we demonstrated that the pre-takeoff controller enables a seamless transition from rolling to flying.

A. Rolling Mode

To illustrate the ground locomotion performance of CAROLLINE, we verified whether it exhibits the commanded motion when a desired velocity V_d^I is provided as input. Using a joystick, we produced velocity commands V_{dx}^I, V_{dy}^I along the e_x^I and e_y^I axes, ensuring that CAROLLINE travels from the START point to the GOAL without deviating from a 1 m-wide path, as shown in Fig. 6. We observed that CAROLLINE can produce the appropriate velocity toward the desired direction. Moreover, we verified that a user can give commands irrespective of the attitude of CAROLLINE's body. Fig. 6 shows the plots of desired and actual velocities of CAROLLINE in the direction of x and y axes. We observed that the actual velocity closely follows the reference command, although small oscillations appear, presumably due to the cage's imperfect spherical structure; further refinements using a more regularly patterned shape would likely reduce them.

Although CAROLLINE's attitude changes continuously while rolling, users do not need to consider it for steering. All they have to do is to check CAROLLINE's position in the inertial frame and provide the desired velocity commands to it. This experiment demonstrated that the sophisticated model

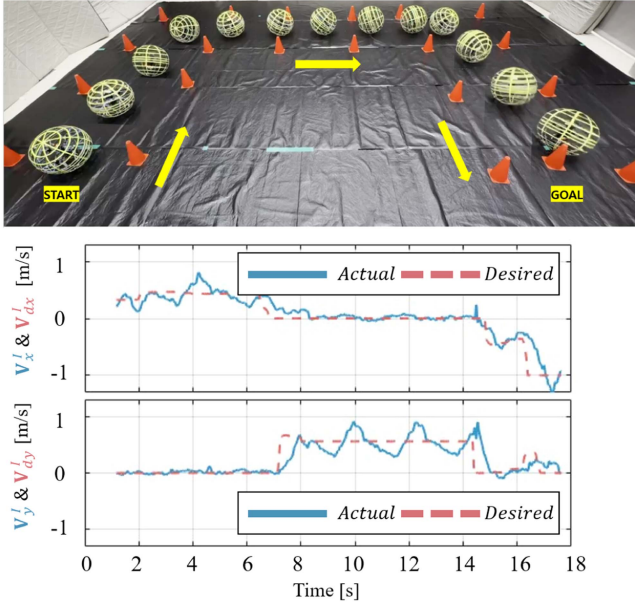


Fig. 6. Experiment in rolling mode: CAROLLINE travels from START to GOAL within a 1 m-wide path while maintaining the desired velocity.

can achieve sufficiently precise control even along a narrow corridor. Additionally, CAROLLINE autonomously determines and generates the required thrust corresponding to given velocity commands, regardless of its current attitude.

B. Pre-Takeoff Mode

To illustrate the attitude-recovery capability of CAROLLINE for a smooth transition from rolling to flying, the performance of the pre-takeoff controller was evaluated across a range of initial attitudes. Specifically, we examined two cases in which CAROLLINE was tilted by 90° and 180° around the e_x axis. Fig. 7 depict CAROLLINE's attitude over time for each initial tilt, showing how it eventually reached a flight-ready orientation. Below each time-series image, we plot the shape of the body frame at the corresponding moment, as well as the tilt angle in radians from the horizontal plane with respect to the e_x axis.

As shown in Fig. 7, the propellers initially faced sideways or downward, but after activating the pre-takeoff mode, they were all directed upwards with respect to the ground. The tilt angles of 90° and 180° took approximately 2.19 and 4.42 s to recover respectively. This confirms that automatic transitioning from rolling to flying can be performed well, making it particularly useful when CAROLLINE's attitude is not immediately clear and a prompt switch to flight from rolling is desired.

C. Energy Efficiency Analysis

We conducted an experiment comparing the energy efficiency of terrestrial and aerial locomotion. Specifically, average power consumption during straight-line motion was measured on a vinyl floor at a low constant speed of 0.5 m/s. To minimize the cage's geometric effects, we took and averaged measurements over 10 m of travel for the test. We repeated the test in aerial locomotion at the same speed and calculated the corresponding

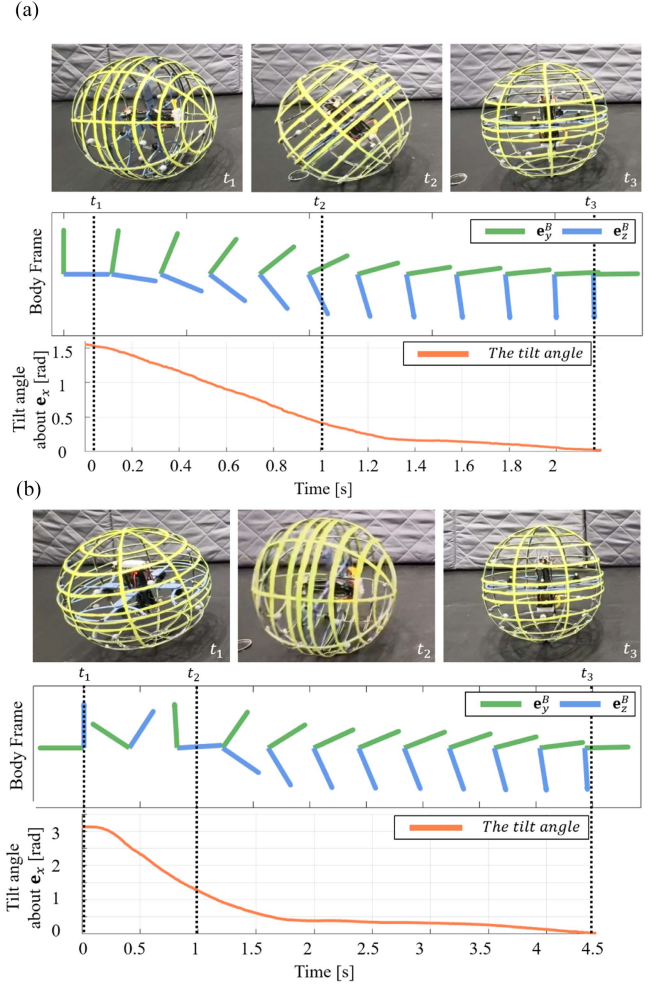


Fig. 7. Pre-takeoff preparation: CAROLLINE's attitude at each time, along with the corresponding body frame and the angle difference from the horizontal plane. (a) Starting at 90° , (b) Starting at 180° .

TABLE II
ENERGY CONSUMPTION AND COT FOR TERRESTRIAL AND AERIAL LOCOMOTION

	Terrestrial Locomotion	Aerial Locomotion
Power (W)	33.2	416.9
COT	7.2	90.7

cost of transport (COT), which quantifies the energy efficiency in transporting a vehicle from one place to another.

As shown in Table II, CAROLLINE's terrestrial mode achieves a COT roughly 12.6 times more efficient than its aerial mode, surpassing HyTAQ's reported ratio of about 10 [13]. Quadrolltor, which has a similar ground-travel approach, reported a ratio of approximately 15 [15], [16]. Such slight degradation may occur because CAROLLINE's negative thrust is generated by spinning the propellers in reverse relative to their designed direction. Although bidirectional propellers can mitigate this, we did not use them for fair comparison and broader applicability. In any case, this experiment highlighted that a minimally modified cage-based rolling design can offer excellent energy efficiency.

VI. CONCLUSION

This letter has presented CAROLLINE, a roadable drone having only a simple spherical cage without complex mechanical transmissions, providing omnidirectional ground locomotion. Despite its structural simplicity, careful modeling and dual-purpose control strategies enable holonomic ground movement and swift reorientation through the pre-takeoff mode. In particular, experiments demonstrated that terrestrial travel is approximately 12.6 times more energy-efficient than flying over the same distance, showing that high efficiency and performance can be simultaneously achieved with no additional actuators. This simple hybrid locomotion design, based on direct coupling with a cage, promises extended operational time and collision safety. It also shows strong potential for practical industrial applications, even in narrow, cluttered indoor settings.

Future efforts will focus on onboard state estimation without motion-capture systems, addressing attitude and position drift through methods like visual-inertial SLAM and ultra wide-band (UWB) localization. This will enable extended outdoor experiments, including energy-efficiency comparisons across a wider range of speeds and terrains, thereby facilitating a more comprehensive evaluation of hybrid locomotion performance. Further improvements will involve complex trajectory tracking, advanced control algorithms that account for torque constraints, and contact-force estimation through observer design, allowing for better torque allocation during aggressive rolling or mode transitions.

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